SoccerGMR Standard Test - GMR / Multi methods

Test value with (delta vs GMR baseline) below | G-mIoU: 0-1 | Other values / deltas: % / percentage points

		Threshold-independent (%)					Threshold = 0.4				Threshold = 0.5				Threshold = 0.6			
Backbone	**Method**	**AUROC**	**mAP**	**mR@1**	**mR@5**	**mR+@5**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**
Moment-DETR	**GMR (baseline)**	**68.80**	**5.91**	**3.68**	**7.98**	**1.16**	**0.00**	**0.00**	**0.0577**	**0.0220**	**100.00**	**75.00**	**0.5239**	**0.4912**	**100.00**	**75.00**	**0.5239**	**0.4912**
	GMR + GEQ Frozen	68.18 **(-0.62)**	5.91 (0.00)	3.68 (0.00)	7.98 (0.00)	1.16 (0.00)	100.00 **(+100.00)**	75.00 **(+75.00)**	0.5239 **(+0.4662)**	0.4912 **(+0.4692)**	100.00 (0.00)	75.00 (0.00)	0.5239 (0.0000)	0.4912 (0.0000)	100.00 (0.00)	75.00 (0.00)	0.5239 (0.0000)	0.4912 (0.0000)
	GMR + GEQ Full	68.96 **(+0.16)**	6.09 **(+0.18)**	3.77 **(+0.09)**	8.19 **(+0.21)**	0.66 **(-0.50)**	99.59 **(+99.59)**	74.81 **(+74.81)**	0.5243 **(+0.4666)**	0.4901 **(+0.4681)**	100.00 (0.00)	75.00 (0.00)	0.5263 **(+0.0024)**	0.4920 **(+0.0008)**	100.00 (0.00)	75.00 (0.00)	0.5263 **(+0.0024)**	0.4920 **(+0.0008)**
	GMR + Multi	70.14 **(+1.34)**	7.27 **(+1.36)**	4.71 **(+1.03)**	10.04 **(+2.06)**	0.28 **(-0.88)**	83.54 **(+83.54)**	69.60 **(+69.60)**	0.4421 **(+0.3844)**	0.4136 **(+0.3916)**	100.00 (0.00)	75.00 (0.00)	0.5188 **(-0.0051)**	0.4895 **(-0.0017)**	100.00 (0.00)	75.00 (0.00)	0.5188 **(-0.0051)**	0.4895 **(-0.0017)**
	GMR + Multi + GEQ Frozen	69.25 **(+0.45)**	7.27 **(+1.36)**	4.71 **(+1.03)**	10.04 **(+2.06)**	0.28 **(-0.88)**	90.65 **(+90.65)**	72.11 **(+72.11)**	0.4749 **(+0.4172)**	0.4456 **(+0.4236)**	100.00 (0.00)	75.00 (0.00)	0.5188 **(-0.0051)**	0.4895 **(-0.0017)**	100.00 (0.00)	75.00 (0.00)	0.5188 **(-0.0051)**	0.4895 **(-0.0017)**
	GMR + Multi + GEQ Full	69.83 **(+1.03)**	7.72 **(+1.81)**	4.06 **(+0.38)**	12.48 **(+4.50)**	0.83 **(-0.33)**	81.91 **(+81.91)**	68.89 **(+68.89)**	0.4395 **(+0.3818)**	0.4070 **(+0.3850)**	100.00 (0.00)	75.00 (0.00)	0.5237 **(-0.0002)**	0.4912 (0.0000)	100.00 (0.00)	75.00 (0.00)	0.5237 **(-0.0002)**	0.4912 (0.0000)
EaTR	**GMR (baseline)**	**76.56**	**16.20**	**10.64**	**21.38**	**9.59**	**48.58**	**57.11**	**0.3332**	**0.2834**	**66.67**	**65.60**	**0.4085**	**0.3600**	**87.20**	**73.46**	**0.4956**	**0.4486**
	GMR + GEQ Frozen	75.17 **(-1.39)**	16.20 (0.00)	10.64 (0.00)	21.38 (0.00)	9.59 (0.00)	63.82 **(+15.24)**	64.02 **(+6.91)**	0.3911 **(+0.0579)**	0.3452 **(+0.0618)**	87.60 **(+20.93)**	72.93 **(+7.33)**	0.4933 **(+0.0848)**	0.4489 **(+0.0889)**	96.75 **(+9.55)**	74.90 **(+1.44)**	0.5315 **(+0.0359)**	0.4858 **(+0.0372)**
	GMR + GEQ Full	75.26 **(-1.30)**	17.15 **(+0.95)**	11.47 **(+0.83)**	22.84 **(+1.46)**	10.79 **(+1.20)**	64.23 **(+15.65)**	64.29 **(+7.18)**	0.3988 **(+0.0656)**	0.3482 **(+0.0648)**	90.65 **(+23.98)**	74.77 **(+9.17)**	0.5111 **(+0.1026)**	0.4635 **(+0.1035)**	98.17 **(+10.98)**	75.23 **(+1.77)**	0.5402 **(+0.0446)**	0.4921 **(+0.0435)**
	GMR + Multi	77.11 **(+0.55)**	17.52 **(+1.32)**	11.64 **(+1.00)**	23.70 **(+2.32)**	11.77 **(+2.18)**	44.92 **(-3.66)**	55.18 **(-1.93)**	0.3305 **(-0.0027)**	0.2702 **(-0.0132)**	61.59 **(-5.08)**	63.79 **(-1.81)**	0.3988 **(-0.0097)**	0.3424 **(-0.0176)**	75.41 **(-11.79)**	69.15 **(-4.31)**	0.4575 **(-0.0381)**	0.4012 **(-0.0474)**
	GMR + Multi + GEQ Frozen	76.69 **(+0.13)**	17.52 **(+1.32)**	11.64 **(+1.00)**	23.70 **(+2.32)**	11.77 **(+2.18)**	69.51 **(+20.93)**	68.13 **(+11.02)**	0.4277 **(+0.0945)**	0.3738 **(+0.0904)**	81.50 **(+14.84)**	71.04 **(+5.44)**	0.4738 **(+0.0653)**	0.4243 **(+0.0643)**	89.63 **(+2.44)**	73.01 **(-0.45)**	0.5058 **(+0.0102)**	0.4571 **(+0.0085)**
	GMR + Multi + GEQ Full	77.27 **(+0.71)**	19.88 **(+3.68)**	14.05 **(+3.41)**	25.31 **(+3.93)**	12.24 **(+2.65)**	68.90 **(+20.33)**	67.60 **(+10.49)**	0.4318 **(+0.0986)**	0.3741 **(+0.0907)**	81.30 **(+14.63)**	71.05 **(+5.45)**	0.4796 **(+0.0711)**	0.4260 **(+0.0660)**	90.65 **(+3.46)**	73.48 **(+0.02)**	0.5165 **(+0.0209)**	0.4632 **(+0.0146)**
QD-DETR	**GMR (baseline)**	**77.66**	**15.08**	**10.04**	**20.11**	**9.62**	**53.46**	**59.84**	**0.3533**	**0.3035**	**90.04**	**73.77**	**0.5058**	**0.4611**	**97.56**	**75.29**	**0.5355**	**0.4901**
	GMR + GEQ Frozen	77.42 **(-0.24)**	15.08 (0.00)	10.04 (0.00)	20.11 (0.00)	9.62 (0.00)	77.24 **(+23.78)**	70.11 **(+10.27)**	0.4559 **(+0.1026)**	0.4070 **(+0.1035)**	92.48 **(+2.44)**	74.47 **(+0.70)**	0.5161 **(+0.0103)**	0.4711 **(+0.0100)**	96.95 **(-0.61)**	74.88 **(-0.41)**	0.5322 **(-0.0033)**	0.4868 **(-0.0033)**
	GMR + GEQ Full	77.66 (0.00)	16.54 **(+1.46)**	11.48 **(+1.44)**	22.09 **(+1.98)**	10.14 **(+0.52)**	80.49 **(+27.03)**	71.35 **(+11.51)**	0.4727 **(+0.1194)**	0.4232 **(+0.1197)**	93.90 **(+3.86)**	74.52 **(+0.75)**	0.5241 **(+0.0183)**	0.4769 **(+0.0158)**	98.17 **(+0.61)**	75.23 **(-0.06)**	0.5419 **(+0.0064)**	0.4942 **(+0.0041)**
	GMR + Multi	75.56 **(-2.10)**	14.67 **(-0.41)**	9.95 **(-0.09)**	19.50 **(-0.61)**	5.60 **(-4.02)**	53.86 **(+0.41)**	58.37 **(-1.47)**	0.3485 **(-0.0048)**	0.2985 **(-0.0050)**	91.67 **(+1.63)**	74.92 **(+1.15)**	0.5157 **(+0.0099)**	0.4690 **(+0.0079)**	96.54 **(-1.02)**	75.28 **(-0.01)**	0.5308 **(-0.0047)**	0.4846 **(-0.0055)**
	GMR + Multi + GEQ Frozen	74.92 **(-2.74)**	14.67 **(-0.41)**	9.95 **(-0.09)**	19.50 **(-0.61)**	5.60 **(-4.02)**	74.59 **(+21.14)**	68.41 **(+8.57)**	0.4380 **(+0.0847)**	0.3917 **(+0.0882)**	94.51 **(+4.47)**	74.82 **(+1.05)**	0.5224 **(+0.0166)**	0.4772 **(+0.0161)**	97.97 **(+0.41)**	74.67 **(-0.62)**	0.5320 **(-0.0035)**	0.4886 **(-0.0015)**
	GMR + Multi + GEQ Full	74.79 **(-2.87)**	13.71 **(-1.37)**	9.21 **(-0.83)**	18.42 **(-1.69)**	6.44 **(-3.18)**	94.31 **(+40.85)**	74.72 **(+14.88)**	0.5194 **(+0.1661)**	0.4756 **(+0.1721)**	98.98 **(+8.94)**	75.15 **(+1.38)**	0.5368 **(+0.0310)**	0.4932 **(+0.0321)**	99.59 **(+2.03)**	74.87 **(-0.42)**	0.5387 **(+0.0032)**	0.4951 **(+0.0050)**
CG-DETR	**GMR (baseline)**	**75.97**	**16.09**	**10.76**	**22.12**	**9.48**	**96.75**	**75.68**	**0.5382**	**0.4890**	**99.19**	**75.84**	**0.5465**	**0.4975**	**99.59**	**75.91**	**0.5484**	**0.4994**
	GMR + GEQ Frozen	76.39 **(+0.42)**	16.09 (0.00)	10.76 (0.00)	22.12 (0.00)	9.48 (0.00)	75.00 **(-21.75)**	69.04 **(-6.64)**	0.4457 **(-0.0925)**	0.3969 **(-0.0921)**	92.48 **(-6.71)**	74.23 **(-1.61)**	0.5189 **(-0.0276)**	0.4698 **(-0.0277)**	99.19 **(-0.41)**	75.95 **(+0.04)**	0.5463 **(-0.0021)**	0.4976 **(-0.0018)**
	GMR + GEQ Full	74.83 **(-1.14)**	16.58 **(+0.49)**	11.12 **(+0.36)**	22.60 **(+0.48)**	10.97 **(+1.49)**	75.00 **(-21.75)**	68.59 **(-7.09)**	0.4439 **(-0.0943)**	0.3933 **(-0.0957)**	97.36 **(-1.83)**	75.55 **(-0.29)**	0.5381 **(-0.0084)**	0.4889 **(-0.0086)**	99.80 **(+0.20)**	75.42 **(-0.49)**	0.5446 **(-0.0038)**	0.4978 **(-0.0016)**
	GMR + Multi	77.40 **(+1.43)**	16.59 **(+0.50)**	11.21 **(+0.45)**	21.77 **(-0.35)**	8.14 **(-1.34)**	46.95 **(-49.80)**	56.20 **(-19.48)**	0.3369 **(-0.2013)**	0.2783 **(-0.2107)**	75.41 **(-23.78)**	69.61 **(-6.23)**	0.4542 **(-0.0923)**	0.4005 **(-0.0970)**	90.45 **(-9.15)**	74.23 **(-1.68)**	0.5167 **(-0.0317)**	0.4641 **(-0.0353)**
	GMR + Multi + GEQ Frozen	75.95 **(-0.02)**	16.59 **(+0.50)**	11.21 **(+0.45)**	21.77 **(-0.35)**	8.14 **(-1.34)**	70.33 **(-26.42)**	67.32 **(-8.36)**	0.4310 **(-0.1072)**	0.3758 **(-0.1132)**	85.16 **(-14.02)**	72.68 **(-3.16)**	0.4938 **(-0.0527)**	0.4417 **(-0.0558)**	92.68 **(-6.91)**	73.91 **(-2.00)**	0.5198 **(-0.0286)**	0.4701 **(-0.0293)**
	GMR + Multi + GEQ Full	75.77 **(-0.20)**	18.59 **(+2.50)**	12.31 **(+1.55)**	24.98 **(+2.86)**	12.13 **(+2.65)**	59.15 **(-37.60)**	62.72 **(-12.96)**	0.3884 **(-0.1498)**	0.3304 **(-0.1586)**	81.30 **(-17.89)**	71.43 **(-4.41)**	0.4784 **(-0.0681)**	0.4246 **(-0.0729)**	93.90 **(-5.69)**	74.04 **(-1.87)**	0.5244 **(-0.0240)**	0.4750 **(-0.0244)**

Delta = method - GMR baseline in the SAME dataset / Test split / protocol / backbone.

Green: gain Red: decrease Gray: unchanged. Deltas use source precision, then round for display. N/A: not applicable.

GEQ checkpoints selected by Val G-mIoU@1 mean at 0.4 / 0.5 / 0.6. Native parent training budgets differ.

N-acc = TN / negative queries; Rej-F1 treats rejection as the target class. Threshold = existence-score threshold.

Source branch: gmr-mr-detr-multiroute-20261030 | Base: 22a882d45eb9 + verified supplemental Test exports

Charades Test-ID - existing methods

Test value with (delta vs GMR baseline) below | G-mIoU: 0-1 | Other values / deltas: % / percentage points

		Threshold-independent (%)					Threshold = 0.4				Threshold = 0.5				Threshold = 0.6			
Backbone	**Method**	**AUROC**	**mAP**	**mR@1**	**mR@5**	**mR+@5**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**
Moment-DETR	**GMR (baseline)**	**84.84**	**30.99**	**25.31**	**38.44**	**N/A**	**76.34**	**76.78**	**0.5625**	**0.4420**	**80.54**	**77.56**	**0.5739**	**0.4598**	**84.49**	**77.50**	**0.5806**	**0.4752**
	GMR + GEQ Frozen	84.50 **(-0.34)**	30.99 (0.00)	25.31 (0.00)	38.44 (0.00)	N/A	71.94 **(-4.41)**	75.42 **(-1.35)**	0.5449 **(-0.0176)**	0.4214 **(-0.0206)**	74.11 **(-6.42)**	75.90 **(-1.66)**	0.5515 **(-0.0224)**	0.4309 **(-0.0289)**	76.59 **(-7.90)**	76.49 **(-1.01)**	0.5585 **(-0.0221)**	0.4414 **(-0.0337)**
	GMR + GEQ Full	85.63 **(+0.79)**	31.12 **(+0.13)**	25.15 **(-0.16)**	38.59 **(+0.15)**	N/A	69.46 **(-6.88)**	75.07 **(-1.70)**	0.5386 **(-0.0239)**	0.4111 **(-0.0309)**	73.06 **(-7.47)**	76.43 **(-1.12)**	0.5517 **(-0.0222)**	0.4275 **(-0.0323)**	76.21 **(-8.28)**	77.22 **(-0.28)**	0.5606 **(-0.0200)**	0.4409 **(-0.0343)**
	GMR + Multi	85.80 **(+0.96)**	31.83 **(+0.84)**	26.14 **(+0.83)**	38.56 **(+0.12)**	N/A	81.29 **(+4.95)**	78.66 **(+1.88)**	0.5885 **(+0.0260)**	0.4671 **(+0.0252)**	83.76 **(+3.23)**	78.83 **(+1.27)**	0.5945 **(+0.0206)**	0.4774 **(+0.0176)**	85.75 **(+1.26)**	78.42 **(+0.92)**	0.5955 **(+0.0149)**	0.4844 **(+0.0092)**
	GMR + Multi + GEQ Frozen	85.83 **(+0.99)**	31.83 **(+0.84)**	26.14 **(+0.83)**	38.56 **(+0.12)**	N/A	80.11 **(+3.76)**	79.15 **(+2.37)**	0.5849 **(+0.0224)**	0.4620 **(+0.0200)**	80.40 **(-0.13)**	79.19 **(+1.63)**	0.5858 **(+0.0119)**	0.4633 **(+0.0035)**	80.75 **(-3.74)**	79.20 **(+1.70)**	0.5866 **(+0.0060)**	0.4647 **(-0.0104)**
	GMR + Multi + GEQ Full	86.51 **(+1.67)**	31.65 **(+0.66)**	25.75 **(+0.44)**	38.55 **(+0.11)**	N/A	74.19 **(-2.15)**	77.56 **(+0.78)**	0.5624 **(-0.0001)**	0.4348 **(-0.0072)**	76.94 **(-3.60)**	78.49 **(+0.93)**	0.5722 **(-0.0017)**	0.4472 **(-0.0126)**	79.11 **(-5.38)**	78.78 **(+1.28)**	0.5794 **(-0.0012)**	0.4568 **(-0.0183)**
EaTR	**GMR (baseline)**	**82.30**	**34.59**	**27.81**	**42.84**	**N/A**	**61.42**	**69.47**	**0.5072**	**0.3738**	**70.32**	**73.08**	**0.5385**	**0.4139**	**79.60**	**75.32**	**0.5663**	**0.4541**
	GMR + GEQ Frozen	82.44 **(+0.14)**	34.59 (0.00)	27.81 (0.00)	42.84 (0.00)	N/A	58.84 **(-2.58)**	68.47 **(-1.00)**	0.4992 **(-0.0080)**	0.3626 **(-0.0113)**	67.45 **(-2.88)**	72.27 **(-0.81)**	0.5304 **(-0.0081)**	0.4016 **(-0.0123)**	76.72 **(-2.88)**	74.64 **(-0.68)**	0.5561 **(-0.0101)**	0.4411 **(-0.0130)**
	GMR + GEQ Full	82.03 **(-0.27)**	34.44 **(-0.15)**	27.69 **(-0.12)**	42.26 **(-0.58)**	N/A	53.66 **(-7.77)**	65.54 **(-3.93)**	0.4785 **(-0.0287)**	0.3383 **(-0.0355)**	62.53 **(-7.80)**	70.20 **(-2.89)**	0.5126 **(-0.0259)**	0.3793 **(-0.0346)**	71.72 **(-7.88)**	73.41 **(-1.92)**	0.5418 **(-0.0245)**	0.4197 **(-0.0344)**
	GMR + Multi	82.76 **(+0.46)**	32.38 **(-2.21)**	26.71 **(-1.10)**	39.24 **(-3.60)**	N/A	59.95 **(-1.48)**	69.48 **(+0.01)**	0.5046 **(-0.0026)**	0.3680 **(-0.0058)**	66.94 **(-3.39)**	72.67 **(-0.41)**	0.5311 **(-0.0075)**	0.4001 **(-0.0138)**	73.60 **(-5.99)**	74.58 **(-0.74)**	0.5523 **(-0.0140)**	0.4294 **(-0.0247)**
	GMR + Multi + GEQ Frozen	82.12 **(-0.18)**	32.38 **(-2.21)**	26.71 **(-1.10)**	39.24 **(-3.60)**	N/A	56.37 **(-5.05)**	67.43 **(-2.05)**	0.4903 **(-0.0169)**	0.3513 **(-0.0225)**	63.17 **(-7.15)**	70.86 **(-2.23)**	0.5152 **(-0.0233)**	0.3823 **(-0.0316)**	70.30 **(-9.30)**	73.07 **(-2.26)**	0.5367 **(-0.0296)**	0.4132 **(-0.0409)**
	GMR + Multi + GEQ Full	82.01 **(-0.29)**	31.25 **(-3.34)**	25.93 **(-1.88)**	37.75 **(-5.09)**	N/A	53.58 **(-7.85)**	66.09 **(-3.38)**	0.4757 **(-0.0315)**	0.3371 **(-0.0367)**	60.24 **(-10.08)**	69.62 **(-3.46)**	0.5006 **(-0.0379)**	0.3677 **(-0.0462)**	66.51 **(-13.09)**	71.92 **(-3.41)**	0.5205 **(-0.0457)**	0.3952 **(-0.0589)**
	GMR + Multi (positive-only)	82.51 **(+0.21)**	33.84 **(-0.75)**	27.92 **(+0.11)**	41.05 **(-1.79)**	N/A	56.26 **(-5.16)**	67.19 **(-2.28)**	0.4901 **(-0.0171)**	0.3509 **(-0.0229)**	65.65 **(-4.68)**	71.62 **(-1.46)**	0.5262 **(-0.0123)**	0.3942 **(-0.0197)**	75.54 **(-4.06)**	74.80 **(-0.52)**	0.5579 **(-0.0084)**	0.4378 **(-0.0163)**
QD-DETR	**GMR (baseline)**	**84.72**	**34.17**	**28.74**	**40.16**	**N/A**	**68.68**	**74.19**	**0.5489**	**0.4119**	**75.24**	**76.15**	**0.5703**	**0.4409**	**80.97**	**77.14**	**0.5853**	**0.4650**
	GMR + GEQ Frozen	84.59 **(-0.13)**	34.17 (0.00)	28.74 (0.00)	40.16 (0.00)	N/A	74.06 **(+5.38)**	76.03 **(+1.84)**	0.5665 **(+0.0176)**	0.4357 **(+0.0238)**	79.01 **(+3.76)**	77.09 **(+0.94)**	0.5812 **(+0.0109)**	0.4571 **(+0.0162)**	83.31 **(+2.34)**	77.41 **(+0.27)**	0.5917 **(+0.0064)**	0.4749 **(+0.0099)**
	GMR + GEQ Full	85.17 **(+0.45)**	34.38 **(+0.21)**	29.14 **(+0.40)**	40.41 **(+0.25)**	N/A	66.64 **(-2.04)**	73.40 **(-0.79)**	0.5415 **(-0.0074)**	0.4026 **(-0.0093)**	72.04 **(-3.20)**	75.51 **(-0.64)**	0.5615 **(-0.0088)**	0.4273 **(-0.0136)**	78.17 **(-2.80)**	77.38 **(+0.24)**	0.5826 **(-0.0027)**	0.4548 **(-0.0102)**
	GMR + Multi	83.77 **(-0.95)**	35.21 **(+1.04)**	29.34 **(+0.60)**	43.61 **(+3.45)**	N/A	66.02 **(-2.66)**	72.43 **(-1.76)**	0.5407 **(-0.0082)**	0.4003 **(-0.0116)**	72.39 **(-2.85)**	74.89 **(-1.26)**	0.5625 **(-0.0078)**	0.4288 **(-0.0121)**	78.44 **(-2.53)**	76.12 **(-1.02)**	0.5790 **(-0.0063)**	0.4545 **(-0.0105)**
	GMR + Multi + GEQ Frozen	83.84 **(-0.88)**	35.21 **(+1.04)**	29.34 **(+0.60)**	43.61 **(+3.45)**	N/A	66.85 **(-1.83)**	73.04 **(-1.15)**	0.5439 **(-0.0050)**	0.4042 **(-0.0078)**	73.20 **(-2.04)**	75.31 **(-0.84)**	0.5654 **(-0.0049)**	0.4325 **(-0.0084)**	79.57 **(-1.40)**	76.75 **(-0.39)**	0.5838 **(-0.0015)**	0.4598 **(-0.0052)**
	GMR + Multi + GEQ Full	84.02 **(-0.70)**	35.57 **(+1.40)**	29.26 **(+0.52)**	44.34 **(+4.18)**	N/A	62.53 **(-6.16)**	71.37 **(-2.82)**	0.5272 **(-0.0217)**	0.3842 **(-0.0278)**	69.68 **(-5.56)**	74.31 **(-1.84)**	0.5528 **(-0.0175)**	0.4165 **(-0.0244)**	76.21 **(-4.76)**	75.97 **(-1.17)**	0.5722 **(-0.0131)**	0.4448 **(-0.0202)**
FlashVTG	**GMR (baseline)**	**86.14**	**41.40**	**31.85**	**54.52**	**N/A**	**73.52**	**77.00**	**0.5807**	**0.4386**	**78.66**	**77.95**	**0.5942**	**0.4602**	**83.76**	**78.19**	**0.6042**	**0.4806**
	GMR + GEQ Frozen	86.06 **(-0.08)**	41.40 (0.00)	31.85 (0.00)	54.52 (0.00)	N/A	72.04 **(-1.48)**	76.44 **(-0.56)**	0.5756 **(-0.0051)**	0.4320 **(-0.0066)**	75.89 **(-2.77)**	77.47 **(-0.48)**	0.5873 **(-0.0069)**	0.4487 **(-0.0115)**	79.70 **(-4.06)**	78.12 **(-0.07)**	0.5967 **(-0.0076)**	0.4646 **(-0.0161)**
	GMR + GEQ Full	86.47 **(+0.33)**	41.70 **(+0.30)**	32.19 **(+0.34)**	55.02 **(+0.50)**	N/A	66.21 **(-7.31)**	74.51 **(-2.49)**	0.5602 **(-0.0205)**	0.4074 **(-0.0312)**	72.23 **(-6.42)**	76.99 **(-0.96)**	0.5816 **(-0.0125)**	0.4346 **(-0.0256)**	77.42 **(-6.34)**	78.18 **(-0.02)**	0.5967 **(-0.0076)**	0.4570 **(-0.0237)**
	GMR + Multi-TQAux	86.47 **(+0.33)**	41.37 **(-0.03)**	32.10 **(+0.25)**	54.16 **(-0.36)**	N/A	75.78 **(+2.26)**	77.83 **(+0.83)**	0.5901 **(+0.0094)**	0.4493 **(+0.0107)**	80.65 **(+1.99)**	78.66 **(+0.70)**	0.6020 **(+0.0079)**	0.4695 **(+0.0093)**	84.97 **(+1.21)**	78.81 **(+0.62)**	0.6102 **(+0.0059)**	0.4866 **(+0.0060)**
	GMR + Multi-TQAux + GEQ Frozen	86.57 **(+0.43)**	41.37 **(-0.03)**	32.10 **(+0.25)**	54.16 **(-0.36)**	N/A	76.24 **(+2.72)**	78.28 **(+1.28)**	0.5940 **(+0.0133)**	0.4521 **(+0.0135)**	77.37 **(-1.29)**	78.46 **(+0.51)**	0.5965 **(+0.0023)**	0.4567 **(-0.0035)**	78.60 **(-5.16)**	78.68 **(+0.48)**	0.5990 **(-0.0052)**	0.4617 **(-0.0189)**
	GMR + Multi-TQAux + GEQ Full	87.47 **(+1.33)**	41.40 (0.00)	32.52 **(+0.67)**	53.65 **(-0.87)**	N/A	71.83 **(-1.69)**	77.52 **(+0.52)**	0.5859 **(+0.0052)**	0.4347 **(-0.0039)**	74.27 **(-4.38)**	78.14 **(+0.19)**	0.5936 **(-0.0006)**	0.4454 **(-0.0148)**	77.63 **(-6.13)**	79.17 **(+0.97)**	0.6052 **(+0.0010)**	0.4605 **(-0.0201)**

Delta = method - GMR baseline in the SAME dataset / Test split / protocol / backbone.

Green: gain Red: decrease Gray: unchanged. Deltas use source precision, then round for display. N/A: not applicable.

24 primary methods plus the previously trained EaTR positive-only variant. Single-GT protocol: mR+@5 is N/A.

N-acc = TN / negative queries; Rej-F1 treats rejection as the target class. Threshold = existence-score threshold.

Source branch: gmr-mr-detr-multiroute-20261030 | Base: 22a882d45eb9 + verified supplemental Test exports

Charades Test-OOD - existing methods

Test value with (delta vs GMR baseline) below | G-mIoU: 0-1 | Other values / deltas: % / percentage points

		Threshold-independent (%)					Threshold = 0.4				Threshold = 0.5				Threshold = 0.6			
Backbone	**Method**	**AUROC**	**mAP**	**mR@1**	**mR@5**	**mR+@5**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**
Moment-DETR	**GMR (baseline)**	**99.99**	**30.99**	**25.31**	**38.44**	**N/A**	**100.00**	**89.88**	**0.6808**	**0.5603**	**100.00**	**88.05**	**0.6712**	**0.5571**	**100.00**	**85.64**	**0.6581**	**0.5527**
	GMR + GEQ Frozen	99.88 **(-0.11)**	30.99 (0.00)	25.31 (0.00)	38.44 (0.00)	N/A	100.00 (0.00)	91.40 **(+1.52)**	0.6852 **(+0.0045)**	0.5617 **(+0.0015)**	100.00 (0.00)	90.42 **(+2.38)**	0.6809 **(+0.0097)**	0.5603 **(+0.0032)**	100.00 (0.00)	89.42 **(+3.79)**	0.6755 **(+0.0174)**	0.5585 **(+0.0058)**
	GMR + GEQ Full	100.00 **(+0.01)**	31.12 **(+0.13)**	25.15 **(-0.16)**	38.59 **(+0.15)**	N/A	100.00 (0.00)	92.77 **(+2.89)**	0.6913 **(+0.0105)**	0.5638 **(+0.0035)**	100.00 (0.00)	91.69 **(+3.65)**	0.6864 **(+0.0152)**	0.5621 **(+0.0051)**	100.00 (0.00)	90.42 **(+4.79)**	0.6796 **(+0.0214)**	0.5599 **(+0.0071)**
	GMR + Multi	100.00 **(+0.01)**	31.83 **(+0.84)**	26.14 **(+0.83)**	38.56 **(+0.12)**	N/A	100.00 (0.00)	88.73 **(-1.15)**	0.6820 **(+0.0013)**	0.5607 **(+0.0004)**	100.00 (0.00)	87.43 **(-0.62)**	0.6757 **(+0.0045)**	0.5586 **(+0.0015)**	100.00 (0.00)	85.85 **(+0.22)**	0.6668 **(+0.0086)**	0.5556 **(+0.0029)**
	GMR + Multi + GEQ Frozen	99.96 **(-0.03)**	31.83 **(+0.84)**	26.14 **(+0.83)**	38.56 **(+0.12)**	N/A	100.00 (0.00)	89.96 **(+0.09)**	0.6844 **(+0.0036)**	0.5615 **(+0.0012)**	100.00 (0.00)	89.82 **(+1.78)**	0.6838 **(+0.0125)**	0.5613 **(+0.0042)**	100.00 (0.00)	89.62 **(+3.98)**	0.6829 **(+0.0247)**	0.5610 **(+0.0082)**
	GMR + Multi + GEQ Full	100.00 **(+0.01)**	31.65 **(+0.66)**	25.75 **(+0.44)**	38.55 **(+0.11)**	N/A	100.00 (0.00)	92.11 **(+2.24)**	0.6914 **(+0.0107)**	0.5638 **(+0.0036)**	100.00 (0.00)	91.28 **(+3.23)**	0.6875 **(+0.0163)**	0.5625 **(+0.0054)**	100.00 (0.00)	90.20 **(+4.57)**	0.6838 **(+0.0257)**	0.5613 **(+0.0086)**
EaTR	**GMR (baseline)**	**100.00**	**34.59**	**27.81**	**42.84**	**N/A**	**100.00**	**92.85**	**0.7001**	**0.5667**	**100.00**	**90.04**	**0.6870**	**0.5623**	**100.00**	**86.30**	**0.6683**	**0.5561**
	GMR + GEQ Frozen	100.00 (0.00)	34.59 (0.00)	27.81 (0.00)	42.84 (0.00)	N/A	100.00 (0.00)	93.88 **(+1.03)**	0.7050 **(+0.0049)**	0.5683 **(+0.0016)**	100.00 (0.00)	91.24 **(+1.20)**	0.6932 **(+0.0063)**	0.5644 **(+0.0021)**	100.00 (0.00)	87.40 **(+1.09)**	0.6726 **(+0.0042)**	0.5575 **(+0.0014)**
	GMR + GEQ Full	100.00 (0.00)	34.44 **(-0.15)**	27.69 **(-0.12)**	42.25 **(-0.59)**	N/A	100.00 (0.00)	95.20 **(+2.35)**	0.7102 **(+0.0101)**	0.5701 **(+0.0034)**	100.00 (0.00)	92.76 **(+2.72)**	0.7000 **(+0.0130)**	0.5667 **(+0.0043)**	100.00 (0.00)	89.41 **(+3.11)**	0.6832 **(+0.0148)**	0.5611 **(+0.0049)**
	GMR + Multi	100.00 (0.00)	32.38 **(-2.21)**	26.71 **(-1.10)**	39.24 **(-3.60)**	N/A	100.00 (0.00)	94.07 **(+1.22)**	0.7049 **(+0.0048)**	0.5683 **(+0.0016)**	100.00 (0.00)	92.05 **(+2.01)**	0.6964 **(+0.0094)**	0.5655 **(+0.0031)**	100.00 (0.00)	89.38 **(+3.08)**	0.6842 **(+0.0159)**	0.5614 **(+0.0053)**
	GMR + Multi + GEQ Frozen	99.99 **(-0.01)**	32.38 **(-2.21)**	26.71 **(-1.10)**	39.24 **(-3.60)**	N/A	100.00 (0.00)	94.86 **(+2.01)**	0.7084 **(+0.0083)**	0.5695 **(+0.0028)**	100.00 (0.00)	92.97 **(+2.93)**	0.6994 **(+0.0124)**	0.5665 **(+0.0041)**	100.00 (0.00)	90.04 **(+3.74)**	0.6852 **(+0.0169)**	0.5617 **(+0.0056)**
	GMR + Multi + GEQ Full	99.99 **(-0.01)**	31.25 **(-3.34)**	25.93 **(-1.88)**	37.75 **(-5.09)**	N/A	100.00 (0.00)	95.90 **(+3.05)**	0.7078 **(+0.0077)**	0.5693 **(+0.0026)**	100.00 (0.00)	93.97 **(+3.94)**	0.6994 **(+0.0124)**	0.5665 **(+0.0041)**	100.00 (0.00)	91.56 **(+5.26)**	0.6880 **(+0.0197)**	0.5627 **(+0.0066)**
	GMR + Multi (positive-only)	100.00 (0.00)	33.84 **(-0.75)**	27.92 **(+0.11)**	41.05 **(-1.79)**	N/A	100.00 (0.00)	94.69 **(+1.84)**	0.7088 **(+0.0087)**	0.5696 **(+0.0029)**	100.00 (0.00)	91.89 **(+1.85)**	0.6980 **(+0.0110)**	0.5660 **(+0.0037)**	100.00 (0.00)	88.33 **(+2.03)**	0.6802 **(+0.0119)**	0.5601 **(+0.0040)**
QD-DETR	**GMR (baseline)**	**100.00**	**34.17**	**28.74**	**40.16**	**N/A**	**100.00**	**92.39**	**0.7055**	**0.5685**	**100.00**	**89.94**	**0.6941**	**0.5647**	**100.00**	**87.35**	**0.6805**	**0.5602**
	GMR + GEQ Frozen	99.91 **(-0.09)**	34.17 (0.00)	28.74 (0.00)	40.16 (0.00)	N/A	100.00 (0.00)	90.60 **(-1.79)**	0.6962 **(-0.0093)**	0.5654 **(-0.0031)**	100.00 (0.00)	88.51 **(-1.43)**	0.6862 **(-0.0079)**	0.5621 **(-0.0026)**	100.00 (0.00)	86.23 **(-1.12)**	0.6752 **(-0.0053)**	0.5584 **(-0.0018)**
	GMR + GEQ Full	99.99 **(-0.01)**	34.38 **(+0.21)**	29.14 **(+0.40)**	40.41 **(+0.25)**	N/A	100.00 (0.00)	93.05 **(+0.66)**	0.7083 **(+0.0028)**	0.5694 **(+0.0009)**	100.00 (0.00)	91.42 **(+1.48)**	0.7013 **(+0.0072)**	0.5671 **(+0.0024)**	100.00 (0.00)	89.34 **(+1.98)**	0.6918 **(+0.0113)**	0.5639 **(+0.0038)**
	GMR + Multi	100.00 (0.00)	35.21 **(+1.04)**	29.34 **(+0.60)**	43.61 **(+3.45)**	N/A	100.00 (0.00)	92.47 **(+0.08)**	0.7106 **(+0.0051)**	0.5702 **(+0.0017)**	100.00 (0.00)	90.52 **(+0.58)**	0.7005 **(+0.0065)**	0.5668 **(+0.0022)**	100.00 (0.00)	87.85 **(+0.50)**	0.6868 **(+0.0063)**	0.5623 **(+0.0021)**
	GMR + Multi + GEQ Frozen	100.00 (0.00)	35.21 **(+1.04)**	29.34 **(+0.60)**	43.61 **(+3.45)**	N/A	100.00 (0.00)	92.50 **(+0.11)**	0.7096 **(+0.0041)**	0.5699 **(+0.0014)**	100.00 (0.00)	90.42 **(+0.48)**	0.6994 **(+0.0054)**	0.5665 **(+0.0018)**	100.00 (0.00)	87.81 **(+0.45)**	0.6859 **(+0.0055)**	0.5620 **(+0.0018)**
	GMR + Multi + GEQ Full	100.00 (0.00)	35.57 **(+1.40)**	29.26 **(+0.52)**	44.34 **(+4.18)**	N/A	100.00 (0.00)	94.03 **(+1.65)**	0.7146 **(+0.0090)**	0.5715 **(+0.0030)**	100.00 (0.00)	91.81 **(+1.86)**	0.7044 **(+0.0103)**	0.5681 **(+0.0034)**	100.00 (0.00)	89.12 **(+1.77)**	0.6911 **(+0.0107)**	0.5637 **(+0.0036)**
FlashVTG	**GMR (baseline)**	**100.00**	**41.40**	**31.85**	**54.52**	**N/A**	**100.00**	**91.98**	**0.7131**	**0.5710**	**100.00**	**89.63**	**0.7009**	**0.5670**	**100.00**	**86.77**	**0.6854**	**0.5618**
	GMR + GEQ Frozen	99.99 **(-0.01)**	41.40 (0.00)	31.85 (0.00)	54.52 (0.00)	N/A	100.00 (0.00)	92.40 **(+0.42)**	0.7154 **(+0.0023)**	0.5718 **(+0.0008)**	100.00 (0.00)	90.90 **(+1.27)**	0.7078 **(+0.0070)**	0.5693 **(+0.0023)**	100.00 (0.00)	89.14 **(+2.37)**	0.6981 **(+0.0127)**	0.5660 **(+0.0042)**
	GMR + GEQ Full	99.99 **(-0.01)**	41.70 **(+0.30)**	32.19 **(+0.34)**	55.02 **(+0.50)**	N/A	100.00 (0.00)	94.56 **(+2.58)**	0.7291 **(+0.0160)**	0.5764 **(+0.0053)**	100.00 (0.00)	92.85 **(+3.22)**	0.7205 **(+0.0196)**	0.5735 **(+0.0065)**	100.00 (0.00)	90.64 **(+3.87)**	0.7096 **(+0.0242)**	0.5699 **(+0.0081)**
	GMR + Multi-TQAux	100.00 (0.00)	41.37 **(-0.03)**	32.10 **(+0.25)**	54.16 **(-0.36)**	N/A	100.00 (0.00)	91.34 **(-0.63)**	0.7112 **(-0.0019)**	0.5704 **(-0.0006)**	100.00 (0.00)	89.12 **(-0.50)**	0.6988 **(-0.0021)**	0.5663 **(-0.0007)**	100.00 (0.00)	86.70 **(-0.07)**	0.6853 **(-0.0001)**	0.5618 (0.0000)
	GMR + Multi-TQAux + GEQ Frozen	100.00 (0.00)	41.37 **(-0.03)**	32.10 **(+0.25)**	54.16 **(-0.36)**	N/A	100.00 (0.00)	91.51 **(-0.46)**	0.7128 **(-0.0003)**	0.5709 **(-0.0001)**	100.00 (0.00)	90.98 **(+1.35)**	0.7096 **(+0.0088)**	0.5699 **(+0.0029)**	100.00 (0.00)	90.41 **(+3.64)**	0.7060 **(+0.0206)**	0.5687 **(+0.0069)**
	GMR + Multi-TQAux + GEQ Full	100.00 (0.00)	41.40 (0.00)	32.52 **(+0.67)**	53.65 **(-0.87)**	N/A	100.00 (0.00)	93.68 **(+1.70)**	0.7268 **(+0.0137)**	0.5756 **(+0.0046)**	100.00 (0.00)	92.66 **(+3.04)**	0.7222 **(+0.0213)**	0.5741 **(+0.0071)**	100.00 (0.00)	91.54 **(+4.76)**	0.7170 **(+0.0316)**	0.5723 **(+0.0105)**

Delta = method - GMR baseline in the SAME dataset / Test split / protocol / backbone.

Green: gain Red: decrease Gray: unchanged. Deltas use source precision, then round for display. N/A: not applicable.

7440 weighted query rows (3720 positive / 3720 negative). Same positive pool as Test-ID; presence results differ.

N-acc = TN / negative queries; Rej-F1 treats rejection as the target class. Threshold = existence-score threshold.

Source branch: gmr-mr-detr-multiroute-20261030 | Base: 22a882d45eb9 + verified supplemental Test exports

SoccerGMR Standard Test - other existing methods

Test value with (delta vs GMR baseline) below | G-mIoU: 0-1 | Other values / deltas: % / percentage points

		Threshold-independent (%)					Threshold = 0.4				Threshold = 0.5				Threshold = 0.6			
Backbone	**Method**	**AUROC**	**mAP**	**mR@1**	**mR@5**	**mR+@5**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**
Moment-DETR	**GMR (baseline)**	**68.80**	**5.91**	**3.68**	**7.98**	**1.16**	**0.00**	**0.00**	**0.0577**	**0.0220**	**100.00**	**75.00**	**0.5239**	**0.4912**	**100.00**	**75.00**	**0.5239**	**0.4912**
	GMR + Shared	69.08 (+0.28)	7.28 (+1.37)	2.95 (-0.73)	12.75 (+4.77)	1.00 (-0.16)	89.23 (+89.23)	72.14 (+72.14)	0.4632 (+0.4055)	0.4377 (+0.4157)	92.68 (-7.32)	73.02 (-1.98)	0.4786 (-0.0453)	0.4532 (-0.0380)	94.31 (-5.69)	73.24 (-1.76)	0.4863 (-0.0376)	0.4608 (-0.0304)
	GMR + Shared + GEQ Frozen	68.63 (-0.17)	7.28 (+1.37)	2.95 (-0.73)	12.75 (+4.77)	1.00 (-0.16)	100.00 (+100.00)	75.00 (+75.00)	0.5128 (+0.4551)	0.4875 (+0.4655)	100.00 (0.00)	75.00 (0.00)	0.5128 (-0.0111)	0.4875 (-0.0037)	100.00 (0.00)	75.00 (0.00)	0.5128 (-0.0111)	0.4875 (-0.0037)
	GMR + Shared + GEQ Full	68.75 (-0.05)	6.92 (+1.01)	3.73 (+0.05)	10.60 (+2.62)	0.57 (-0.59)	94.72 (+94.72)	73.73 (+73.73)	0.4982 (+0.4405)	0.4662 (+0.4442)	100.00 (0.00)	75.00 (0.00)	0.5231 (-0.0008)	0.4910 (-0.0002)	100.00 (0.00)	75.00 (0.00)	0.5231 (-0.0008)	0.4910 (-0.0002)
	GMR + SepFFN	69.92 (+1.12)	7.51 (+1.60)	4.76 (+1.08)	10.20 (+2.22)	1.05 (-0.11)	100.00 (+100.00)	75.00 (+75.00)	0.5217 (+0.4640)	0.4905 (+0.4685)	100.00 (0.00)	75.00 (0.00)	0.5217 (-0.0022)	0.4905 (-0.0007)	100.00 (0.00)	75.00 (0.00)	0.5217 (-0.0022)	0.4905 (-0.0007)
	GMR + Instruction	70.33 (+1.53)	5.81 (-0.10)	3.85 (+0.17)	8.21 (+0.23)	0.97 (-0.19)	72.97 (+72.97)	65.93 (+65.93)	0.3947 (+0.3370)	0.3646 (+0.3426)	100.00 (0.00)	75.00 (0.00)	0.5214 (-0.0025)	0.4904 (-0.0008)	100.00 (0.00)	75.00 (0.00)	0.5214 (-0.0025)	0.4904 (-0.0008)
	GMR + Adaptive	68.58 (-0.22)	7.09 (+1.18)	4.38 (+0.70)	10.00 (+2.02)	1.47 (+0.31)	0.00 (0.00)	0.00 (0.00)	0.0620 (+0.0043)	0.0268 (+0.0048)	100.00 (0.00)	75.00 (0.00)	0.5218 (-0.0021)	0.4905 (-0.0007)	100.00 (0.00)	75.00 (0.00)	0.5218 (-0.0021)	0.4905 (-0.0007)
EaTR	**GMR (baseline)**	**76.56**	**16.20**	**10.64**	**21.38**	**9.59**	**48.58**	**57.11**	**0.3332**	**0.2834**	**66.67**	**65.60**	**0.4085**	**0.3600**	**87.20**	**73.46**	**0.4956**	**0.4486**
	GMR + Shared	77.57 (+1.01)	16.65 (+0.45)	11.20 (+0.56)	23.55 (+2.17)	8.60 (-0.99)	55.69 (+7.11)	61.23 (+4.12)	0.3682 (+0.0350)	0.3139 (+0.0305)	65.45 (-1.22)	65.92 (+0.32)	0.4085 (0.0000)	0.3551 (-0.0049)	74.59 (-12.60)	69.84 (-3.62)	0.4473 (-0.0483)	0.3953 (-0.0533)
	GMR + Shared + GEQ Frozen	76.58 (+0.02)	16.65 (+0.45)	11.20 (+0.56)	23.55 (+2.17)	8.60 (-0.99)	60.37 (+11.79)	62.86 (+5.75)	0.3813 (+0.0481)	0.3304 (+0.0470)	72.76 (+6.10)	68.13 (+2.53)	0.4333 (+0.0248)	0.3847 (+0.0247)	82.32 (-4.88)	71.55 (-1.91)	0.4723 (-0.0233)	0.4259 (-0.0227)
	GMR + Shared + GEQ Full	76.26 (-0.30)	17.15 (+0.95)	11.84 (+1.20)	23.36 (+1.98)	9.48 (-0.11)	56.30 (+7.72)	61.42 (+4.31)	0.3707 (+0.0375)	0.3163 (+0.0329)	69.92 (+3.25)	67.12 (+1.52)	0.4247 (+0.0162)	0.3742 (+0.0142)	79.47 (-7.72)	70.64 (-2.82)	0.4660 (-0.0296)	0.4165 (-0.0321)
	GMR + SepFFN	77.75 (+1.19)	17.06 (+0.86)	11.32 (+0.68)	22.59 (+1.21)	9.21 (-0.38)	44.72 (-3.86)	54.93 (-2.18)	0.3243 (-0.0089)	0.2655 (-0.0179)	66.87 (+0.20)	66.87 (+1.27)	0.4126 (+0.0041)	0.3615 (+0.0015)	86.18 (-1.02)	72.91 (-0.55)	0.4901 (-0.0055)	0.4425 (-0.0061)
	GMR + Instruction	77.02 (+0.46)	17.48 (+1.28)	11.52 (+0.88)	23.82 (+2.44)	11.14 (+1.55)	56.71 (+8.13)	61.39 (+4.28)	0.3750 (+0.0418)	0.3183 (+0.0349)	69.72 (+3.05)	67.25 (+1.65)	0.4272 (+0.0187)	0.3741 (+0.0141)	83.54 (-3.66)	72.36 (-1.10)	0.4830 (-0.0126)	0.4339 (-0.0147)
	GMR + Adaptive	78.01 (+1.45)	17.29 (+1.09)	11.56 (+0.92)	22.99 (+1.61)	10.32 (+0.73)	63.41 (+14.84)	65.27 (+8.16)	0.4050 (+0.0718)	0.3513 (+0.0679)	81.71 (+15.04)	71.98 (+6.38)	0.4772 (+0.0687)	0.4293 (+0.0693)	91.46 (+4.27)	74.32 (+0.86)	0.5160 (+0.0204)	0.4697 (+0.0211)
QD-DETR	**GMR (baseline)**	**77.66**	**15.08**	**10.04**	**20.11**	**9.62**	**53.46**	**59.84**	**0.3533**	**0.3035**	**90.04**	**73.77**	**0.5058**	**0.4611**	**97.56**	**75.29**	**0.5355**	**0.4901**
	GMR + Shared	77.13 (-0.53)	15.24 (+0.16)	10.64 (+0.60)	20.00 (-0.11)	9.78 (+0.16)	59.96 (+6.50)	63.37 (+3.53)	0.3844 (+0.0311)	0.3334 (+0.0299)	72.76 (-17.28)	68.78 (-4.99)	0.4392 (-0.0666)	0.3887 (-0.0724)	82.52 (-15.04)	71.86 (-3.43)	0.4795 (-0.0560)	0.4308 (-0.0593)
	GMR + Shared + GEQ Frozen	76.29 (-1.37)	15.24 (+0.16)	10.64 (+0.60)	20.00 (-0.11)	9.78 (+0.16)	65.65 (+12.20)	65.06 (+5.22)	0.4056 (+0.0523)	0.3557 (+0.0522)	73.78 (-16.26)	68.23 (-5.54)	0.4379 (-0.0679)	0.3896 (-0.0715)	83.94 (-13.62)	72.08 (-3.21)	0.4817 (-0.0538)	0.4342 (-0.0559)
	GMR + Shared + GEQ Full	76.31 (-1.35)	15.69 (+0.61)	10.71 (+0.67)	20.15 (+0.04)	10.42 (+0.80)	66.46 (+13.01)	65.01 (+5.17)	0.4091 (+0.0558)	0.3596 (+0.0561)	74.80 (-15.24)	68.40 (-5.37)	0.4441 (-0.0617)	0.3957 (-0.0654)	84.35 (-13.21)	72.36 (-2.93)	0.4866 (-0.0489)	0.4382 (-0.0519)
	GMR + SepFFN	75.57 (-2.09)	11.03 (-4.05)	7.34 (-2.70)	15.36 (-4.75)	4.16 (-5.46)	72.97 (+19.51)	68.25 (+8.41)	0.4185 (+0.0652)	0.3783 (+0.0748)	83.74 (-6.30)	71.78 (-1.99)	0.4642 (-0.0416)	0.4256 (-0.0355)	93.09 (-4.47)	74.11 (-1.18)	0.5026 (-0.0329)	0.4653 (-0.0248)
	GMR + Instruction	76.25 (-1.41)	14.46 (-0.62)	9.99 (-0.05)	19.26 (-0.85)	6.84 (-2.78)	54.07 (+0.61)	60.18 (+0.34)	0.3515 (-0.0018)	0.3035 (0.0000)	79.27 (-10.77)	70.52 (-3.25)	0.4601 (-0.0457)	0.4138 (-0.0473)	90.24 (-7.32)	74.00 (-1.29)	0.5064 (-0.0291)	0.4601 (-0.0300)
	GMR + Adaptive	75.47 (-2.19)	14.68 (-0.40)	9.46 (-0.58)	20.46 (+0.35)	6.91 (-2.71)	43.29 (-10.16)	52.66 (-7.18)	0.2988 (-0.0545)	0.2485 (-0.0550)	65.04 (-25.00)	64.65 (-9.12)	0.3918 (-0.1140)	0.3432 (-0.1179)	88.82 (-8.74)	73.32 (-1.97)	0.4952 (-0.0403)	0.4505 (-0.0396)
CG-DETR	**GMR (baseline)**	**75.97**	**16.09**	**10.76**	**22.12**	**9.48**	**96.75**	**75.68**	**0.5382**	**0.4890**	**99.19**	**75.84**	**0.5465**	**0.4975**	**99.59**	**75.91**	**0.5484**	**0.4994**
	GMR + Shared	77.44 (+1.47)	15.09 (-1.00)	9.92 (-0.84)	20.57 (-1.55)	9.66 (+0.18)	39.23 (-57.52)	50.20 (-25.48)	0.2883 (-0.2499)	0.2379 (-0.2511)	69.51 (-29.67)	67.72 (-8.12)	0.4189 (-0.1276)	0.3725 (-0.1250)	88.21 (-11.38)	73.68 (-2.23)	0.4965 (-0.0519)	0.4507 (-0.0487)
	GMR + Shared + GEQ Frozen	77.32 (+1.35)	15.09 (-1.00)	9.92 (-0.84)	20.57 (-1.55)	9.66 (+0.18)	70.12 (-26.63)	67.32 (-8.36)	0.4184 (-0.1198)	0.3722 (-0.1168)	85.37 (-13.82)	72.41 (-3.43)	0.4834 (-0.0631)	0.4382 (-0.0593)	93.50 (-6.10)	74.19 (-1.72)	0.5151 (-0.0333)	0.4708 (-0.0286)
	GMR + Shared + GEQ Full	75.79 (-0.18)	14.22 (-1.87)	9.28 (-1.48)	19.39 (-2.73)	9.22 (-0.26)	74.59 (-22.15)	68.34 (-7.34)	0.4318 (-0.1064)	0.3891 (-0.0999)	92.68 (-6.50)	74.33 (-1.51)	0.5113 (-0.0352)	0.4667 (-0.0308)	97.15 (-2.44)	75.04 (-0.87)	0.5288 (-0.0196)	0.4846 (-0.0148)
	GMR + SepFFN	75.52 (-0.45)	15.04 (-1.05)	10.65 (-0.11)	20.22 (-1.90)	4.62 (-4.86)	53.46 (-43.29)	59.37 (-16.31)	0.3560 (-0.1822)	0.3022 (-0.1868)	91.06 (-8.13)	73.87 (-1.97)	0.5133 (-0.0332)	0.4632 (-0.0343)	98.17 (-1.42)	75.71 (-0.20)	0.5437 (-0.0047)	0.4936 (-0.0058)
	GMR + Instruction	76.96 (+0.99)	15.67 (-0.42)	10.72 (-0.04)	21.34 (-0.78)	8.46 (-1.02)	25.81 (-70.93)	38.60 (-37.08)	0.2360 (-0.3022)	0.1806 (-0.3084)	57.93 (-41.26)	62.16 (-13.68)	0.3733 (-0.1732)	0.3210 (-0.1765)	89.02 (-10.57)	73.24 (-2.67)	0.5033 (-0.0451)	0.4544 (-0.0450)
	GMR + Adaptive	77.27 (+1.30)	18.79 (+2.70)	12.89 (+2.13)	24.39 (+2.27)	13.06 (+3.58)	64.84 (-31.91)	65.10 (-10.58)	0.4107 (-0.1275)	0.3579 (-0.1311)	87.60 (-11.59)	73.42 (-2.42)	0.5045 (-0.0420)	0.4531 (-0.0444)	96.95 (-2.64)	75.89 (-0.02)	0.5404 (-0.0080)	0.4915 (-0.0079)

Delta = method - GMR baseline in the SAME dataset / Test split / protocol / backbone.

Green: gain Red: decrease Gray: unchanged. Deltas use source precision, then round for display. N/A: not applicable.

Includes the completed EaTR Adaptive Test omitted from the old branch snapshot. Missing threshold reports supplemented from formal predictions.

N-acc = TN / negative queries; Rej-F1 treats rejection as the target class. Threshold = existence-score threshold.

Source branch: gmr-mr-detr-multiroute-20261030 | Base: 22a882d45eb9 + verified supplemental Test exports

SoccerGMR Standard Test - historical main E1 initialization

Test value with (delta vs GMR baseline) below | G-mIoU: 0-1 | Other values / deltas: % / percentage points

		Threshold-independent (%)					Threshold = 0.4				Threshold = 0.5				Threshold = 0.6			
Backbone	**Method**	**AUROC**	**mAP**	**mR@1**	**mR@5**	**mR+@5**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**
Moment-DETR	**GMR (baseline)**	**70.20**	**6.68**	**4.64**	**9.44**	**0.88**	**0.00**	**0.00**	**0.0583**	**0.0243**	**79.27**	**67.94**	**0.4250**	**0.3939**	**100.00**	**74.94**	**0.5202**	**0.4900**
	GMR + GEQ Frozen	69.36 **(-0.84)**	6.68 (0.00)	4.64 (0.00)	9.44 (0.00)	0.88 (0.00)	97.76 **(+97.76)**	74.23 **(+74.23)**	0.5099 **(+0.4516)**	0.4796 **(+0.4553)**	100.00 **(+20.73)**	75.00 **(+7.06)**	0.5202 **(+0.0952)**	0.4900 **(+0.0961)**	100.00 (0.00)	75.00 **(+0.06)**	0.5202 (0.0000)	0.4900 (0.0000)
	GMR + GEQ Full	70.21 **(+0.01)**	5.77 **(-0.91)**	3.64 **(-1.00)**	8.36 **(-1.08)**	1.23 **(+0.35)**	89.02 **(+89.02)**	71.80 **(+71.80)**	0.4699 **(+0.4116)**	0.4395 **(+0.4152)**	100.00 **(+20.73)**	75.00 **(+7.06)**	0.5211 **(+0.0961)**	0.4903 **(+0.0964)**	100.00 (0.00)	75.00 **(+0.06)**	0.5211 **(+0.0009)**	0.4903 **(+0.0003)**
EaTR	**GMR (baseline)**	**76.56**	**16.49**	**10.86**	**21.73**	**9.59**	**48.58**	**57.11**	**0.3341**	**0.2837**	**66.67**	**65.60**	**0.4094**	**0.3604**	**87.20**	**73.46**	**0.4966**	**0.4489**
	GMR + GEQ Frozen	75.17 **(-1.39)**	16.49 (0.00)	10.86 (0.00)	21.73 (0.00)	9.59 (0.00)	63.82 **(+15.24)**	64.02 **(+6.91)**	0.3921 **(+0.0580)**	0.3455 **(+0.0618)**	87.60 **(+20.93)**	72.93 **(+7.33)**	0.4943 **(+0.0849)**	0.4492 **(+0.0888)**	96.75 **(+9.55)**	74.90 **(+1.44)**	0.5324 **(+0.0358)**	0.4861 **(+0.0372)**
	GMR + GEQ Full	75.26 **(-1.30)**	17.48 **(+0.99)**	11.69 **(+0.83)**	23.30 **(+1.57)**	10.79 **(+1.20)**	64.23 **(+15.65)**	64.29 **(+7.18)**	0.3995 **(+0.0654)**	0.3484 **(+0.0647)**	90.65 **(+23.98)**	74.77 **(+9.17)**	0.5119 **(+0.1025)**	0.4638 **(+0.1034)**	98.17 **(+10.98)**	75.23 **(+1.77)**	0.5409 **(+0.0443)**	0.4924 **(+0.0435)**
QD-DETR	**GMR (baseline)**	**77.01**	**14.62**	**9.47**	**20.17**	**7.43**	**56.71**	**61.73**	**0.3651**	**0.3147**	**79.47**	**71.48**	**0.4649**	**0.4147**	**90.65**	**73.29**	**0.5068**	**0.4604**
	GMR + GEQ Frozen	76.32 **(-0.69)**	14.62 (0.00)	9.47 (0.00)	20.17 (0.00)	7.43 (0.00)	76.02 **(+19.31)**	69.26 **(+7.53)**	0.4463 **(+0.0812)**	0.3986 **(+0.0839)**	90.04 **(+10.57)**	73.04 **(+1.56)**	0.5029 **(+0.0380)**	0.4572 **(+0.0425)**	97.15 **(+6.50)**	75.28 **(+1.99)**	0.5339 **(+0.0271)**	0.4878 **(+0.0274)**
	GMR + GEQ Full	75.98 **(-1.03)**	14.83 **(+0.21)**	10.10 **(+0.63)**	19.44 **(-0.73)**	8.30 **(+0.87)**	84.96 **(+28.25)**	71.76 **(+10.03)**	0.4841 **(+0.1190)**	0.4379 **(+0.1232)**	95.12 **(+15.65)**	74.34 **(+2.86)**	0.5255 **(+0.0606)**	0.4806 **(+0.0659)**	98.78 **(+8.13)**	75.06 **(+1.77)**	0.5395 **(+0.0327)**	0.4945 **(+0.0341)**
CG-DETR	**GMR (baseline)**	**76.40**	**18.65**	**12.65**	**24.90**	**11.98**	**23.17**	**35.46**	**0.2284**	**0.1707**	**50.41**	**57.47**	**0.3435**	**0.2914**	**78.46**	**71.09**	**0.4666**	**0.4153**
	GMR + GEQ Frozen	74.78 **(-1.62)**	18.65 (0.00)	12.65 (0.00)	24.90 (0.00)	11.98 (0.00)	77.24 **(+54.07)**	69.34 **(+33.88)**	0.4512 **(+0.2228)**	0.4016 **(+0.2309)**	93.70 **(+43.29)**	74.24 **(+16.77)**	0.5205 **(+0.1770)**	0.4734 **(+0.1820)**	99.19 **(+20.73)**	75.43 **(+4.34)**	0.5428 **(+0.0762)**	0.4958 **(+0.0805)**
	GMR + GEQ Full	74.51 **(-1.89)**	18.71 **(+0.06)**	12.75 **(+0.10)**	23.98 **(-0.92)**	10.41 **(-1.57)**	63.82 **(+40.65)**	64.02 **(+28.56)**	0.4001 **(+0.1717)**	0.3489 **(+0.1782)**	90.85 **(+40.45)**	73.52 **(+16.05)**	0.5110 **(+0.1675)**	0.4618 **(+0.1704)**	98.98 **(+20.53)**	75.21 **(+4.12)**	0.5444 **(+0.0778)**	0.4956 **(+0.0803)**

Delta = method - GMR baseline in the SAME dataset / Test split / protocol / backbone.

Green: gain Red: decrease Gray: unchanged. Deltas use source precision, then round for display. N/A: not applicable.

Historical parent checkpoints. Compare within this table; do not mix their GMR values with the current protocol.

N-acc = TN / negative queries; Rej-F1 treats rejection as the target class. Threshold = existence-score threshold.

Source branch: gmr-mr-detr-multiroute-20261030 | Base: 22a882d45eb9 + verified supplemental Test exports

SoccerGMR Standard Test - historical fixed-four-epoch GEQ

Test value with (delta vs GMR baseline) below | G-mIoU: 0-1 | Other values / deltas: % / percentage points

		Threshold-independent (%)					Threshold = 0.4				Threshold = 0.5				Threshold = 0.6			
Backbone	**Method**	**AUROC**	**mAP**	**mR@1**	**mR@5**	**mR+@5**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**
Moment-DETR	**GMR (baseline)**	**68.80**	**5.91**	**3.68**	**7.98**	**1.16**	**0.00**	**0.00**	**0.0577**	**0.0220**	**100.00**	**75.00**	**0.5239**	**0.4912**	**100.00**	**75.00**	**0.5239**	**0.4912**
	GMR + GEQ Frozen	68.45 **(-0.35)**	5.91 (0.00)	3.68 (0.00)	7.98 (0.00)	1.16 (0.00)	9.76 **(+9.76)**	16.55 **(+16.55)**	0.1018 **(+0.0441)**	0.0672 **(+0.0452)**	99.80 **(-0.20)**	74.90 **(-0.10)**	0.5230 **(-0.0009)**	0.4903 **(-0.0009)**	100.00 (0.00)	75.00 (0.00)	0.5239 (0.0000)	0.4912 (0.0000)
	GMR + Multi + GEQ Frozen	69.12 **(+0.32)**	7.27 **(+1.36)**	4.71 **(+1.03)**	10.04 **(+2.06)**	0.28 **(-0.88)**	37.60 **(+37.60)**	46.02 **(+46.02)**	0.2327 **(+0.1750)**	0.2012 **(+0.1792)**	99.39 **(-0.61)**	74.71 **(-0.29)**	0.5159 **(-0.0080)**	0.4866 **(-0.0046)**	100.00 (0.00)	75.00 (0.00)	0.5188 **(-0.0051)**	0.4895 **(-0.0017)**
	GMR + Shared + GEQ Frozen	68.30 **(-0.50)**	7.28 **(+1.37)**	2.95 **(-0.73)**	12.75 **(+4.77)**	1.00 **(-0.16)**	20.53 **(+20.53)**	30.28 **(+30.28)**	0.1428 **(+0.0851)**	0.1165 **(+0.0945)**	100.00 (0.00)	75.00 (0.00)	0.5128 **(-0.0111)**	0.4875 **(-0.0037)**	100.00 (0.00)	75.00 (0.00)	0.5128 **(-0.0111)**	0.4875 **(-0.0037)**
EaTR	**GMR (baseline)**	**76.56**	**16.20**	**10.64**	**21.38**	**9.59**	**48.58**	**57.11**	**0.3332**	**0.2834**	**66.67**	**65.60**	**0.4085**	**0.3600**	**87.20**	**73.46**	**0.4956**	**0.4486**
	GMR + GEQ Frozen	75.17 **(-1.39)**	16.20 (0.00)	10.64 (0.00)	21.38 (0.00)	9.59 (0.00)	63.82 **(+15.24)**	64.02 **(+6.91)**	0.3911 **(+0.0579)**	0.3452 **(+0.0618)**	87.60 **(+20.93)**	72.93 **(+7.33)**	0.4933 **(+0.0848)**	0.4489 **(+0.0889)**	96.75 **(+9.55)**	74.90 **(+1.44)**	0.5315 **(+0.0359)**	0.4858 **(+0.0372)**
	GMR + Multi + GEQ Frozen	76.69 **(+0.13)**	17.52 **(+1.32)**	11.64 **(+1.00)**	23.70 **(+2.32)**	11.77 **(+2.18)**	69.51 **(+20.93)**	68.13 **(+11.02)**	0.4277 **(+0.0945)**	0.3738 **(+0.0904)**	81.50 **(+14.84)**	71.04 **(+5.44)**	0.4738 **(+0.0653)**	0.4243 **(+0.0643)**	89.63 **(+2.44)**	73.01 **(-0.45)**	0.5058 **(+0.0102)**	0.4571 **(+0.0085)**
	GMR + Shared + GEQ Frozen	76.49 **(-0.07)**	16.65 **(+0.45)**	11.20 **(+0.56)**	23.55 **(+2.17)**	8.60 **(-0.99)**	62.40 **(+13.82)**	64.50 **(+7.39)**	0.3882 **(+0.0550)**	0.3383 **(+0.0549)**	73.37 **(+6.71)**	68.63 **(+3.03)**	0.4351 **(+0.0266)**	0.3864 **(+0.0264)**	81.30 **(-5.89)**	70.55 **(-2.91)**	0.4673 **(-0.0283)**	0.4205 **(-0.0281)**
QD-DETR	**GMR (baseline)**	**77.66**	**15.08**	**10.04**	**20.11**	**9.62**	**53.46**	**59.84**	**0.3533**	**0.3035**	**90.04**	**73.77**	**0.5058**	**0.4611**	**97.56**	**75.29**	**0.5355**	**0.4901**
	GMR + GEQ Frozen	77.42 **(-0.24)**	15.08 (0.00)	10.04 (0.00)	20.11 (0.00)	9.62 (0.00)	77.24 **(+23.78)**	70.11 **(+10.27)**	0.4559 **(+0.1026)**	0.4070 **(+0.1035)**	92.48 **(+2.44)**	74.47 **(+0.70)**	0.5161 **(+0.0103)**	0.4711 **(+0.0100)**	96.95 **(-0.61)**	74.88 **(-0.41)**	0.5322 **(-0.0033)**	0.4868 **(-0.0033)**
	GMR + Multi + GEQ Frozen	74.92 **(-2.74)**	14.67 **(-0.41)**	9.95 **(-0.09)**	19.50 **(-0.61)**	5.60 **(-4.02)**	74.59 **(+21.14)**	68.41 **(+8.57)**	0.4380 **(+0.0847)**	0.3917 **(+0.0882)**	94.51 **(+4.47)**	74.82 **(+1.05)**	0.5224 **(+0.0166)**	0.4772 **(+0.0161)**	97.97 **(+0.41)**	74.67 **(-0.62)**	0.5320 **(-0.0035)**	0.4886 **(-0.0015)**
	GMR + Shared + GEQ Frozen	76.29 **(-1.37)**	15.24 **(+0.16)**	10.64 **(+0.60)**	20.00 **(-0.11)**	9.78 **(+0.16)**	65.65 **(+12.20)**	65.06 **(+5.22)**	0.4056 **(+0.0523)**	0.3557 **(+0.0522)**	73.78 **(-16.26)**	68.23 **(-5.54)**	0.4379 **(-0.0679)**	0.3896 **(-0.0715)**	83.94 **(-13.62)**	72.08 **(-3.21)**	0.4817 **(-0.0538)**	0.4342 **(-0.0559)**
CG-DETR	**GMR (baseline)**	**75.97**	**16.09**	**10.76**	**22.12**	**9.48**	**96.75**	**75.68**	**0.5382**	**0.4890**	**99.19**	**75.84**	**0.5465**	**0.4975**	**99.59**	**75.91**	**0.5484**	**0.4994**
	GMR + GEQ Frozen	76.39 **(+0.42)**	16.09 (0.00)	10.76 (0.00)	22.12 (0.00)	9.48 (0.00)	75.00 **(-21.75)**	69.04 **(-6.64)**	0.4457 **(-0.0925)**	0.3969 **(-0.0921)**	92.48 **(-6.71)**	74.23 **(-1.61)**	0.5189 **(-0.0276)**	0.4698 **(-0.0277)**	99.19 **(-0.41)**	75.95 **(+0.04)**	0.5463 **(-0.0021)**	0.4976 **(-0.0018)**
	GMR + Multi + GEQ Frozen	77.32 **(+1.35)**	16.59 **(+0.50)**	11.21 **(+0.45)**	21.77 **(-0.35)**	8.14 **(-1.34)**	61.38 **(-35.37)**	63.98 **(-11.70)**	0.3971 **(-0.1411)**	0.3401 **(-0.1489)**	79.07 **(-20.12)**	71.18 **(-4.66)**	0.4705 **(-0.0760)**	0.4173 **(-0.0802)**	89.43 **(-10.16)**	73.58 **(-2.33)**	0.5109 **(-0.0375)**	0.4592 **(-0.0402)**
	GMR + Shared + GEQ Frozen	77.32 **(+1.35)**	15.09 **(-1.00)**	9.92 **(-0.84)**	20.57 **(-1.55)**	9.66 **(+0.18)**	70.12 **(-26.63)**	67.32 **(-8.36)**	0.4184 **(-0.1198)**	0.3722 **(-0.1168)**	85.37 **(-13.82)**	72.41 **(-3.43)**	0.4834 **(-0.0631)**	0.4382 **(-0.0593)**	93.50 **(-6.10)**	74.19 **(-1.72)**	0.5151 **(-0.0333)**	0.4708 **(-0.0286)**

Delta = method - GMR baseline in the SAME dataset / Test split / protocol / backbone.

Green: gain Red: decrease Gray: unchanged. Deltas use source precision, then round for display. N/A: not applicable.

Historical frozen GEQ after four fixed epochs, without the newer Val checkpoint-selection protocol.

N-acc = TN / negative queries; Rej-F1 treats rejection as the target class. Threshold = existence-score threshold.

Source branch: gmr-mr-detr-multiroute-20261030 | Base: 22a882d45eb9 + verified supplemental Test exports